Final project Proposal

Title: Predicting FIFA Player Ratings Using Key Attributes

Raja Prabakaran
Nitheesh Samiappan
Pooja Venugopal Baskaran
Aravindh Tiruchirapalli Seetharaman

1. Background and Motivation

Predicting player ratings in EA Sports FIFA based on key attributes is important for both game developers and sports analysts. These accurate player ratings give a better gaming experience and allows users to make right decisions in both the gaming community and for professional sports analysis. Even players of the game may refer to this data to assess the potential of players before signing them in the game. Understanding which attributes most influence a player's overall rating may help create more realistic assessments that can improve gameplay.

2. Research Questions

- How do attributes (dribbling, speed, passing, etc.) affect a player's overall rating?
- Can we predict a player's overall rating using these attributes?
- Which attributes are the best predictors of the overall rating, and how can these be used for more accurate player assessments?

3. Data Description

- Source: Kaggle FIFA Player Dataset (18,000 players from FC24 version).
- The original dataset contains over 180,000 entries from multiple versions of the game, but we have filtered it to include only players from the most recent version, FC24.
- The dataset contains over 108 attributes, and we have selected key attributes that contribute to predicting a player's overall rating, such as:
    - Player positions: Roles in both club and national teams.

    - Player attributes: Categories such as Attacking, Skills, Defense, Mentality, and GK Skills.
    - Player personal data: Nationality, club, date of birth, wage, salary, etc.
- Response variable: The overall player rating.
- Predictor variables: Selected player attributes such as dribbling, passing, and speed.
- Missing Data: For players with missing data (e.g., free agents without a club), we will either remove them or add values based on similar player attributes.

4. Methodology

1. Multiple Linear Regression:
   This technique will be used to predict overall player ratings based on attributes like dribbling, passing, and speed. It's appropriate because it helps establish the relationship between multiple independent variables and the dependent variable (overall rating).

2. Interaction Models:
   Interaction terms will be included to get how combinations of attributes (e.g., dribbling and speed) affect player ratings. This will be useful when the effect of one attribute depends on another.

3. Second-Order Models:
   Second-order terms will be used to model non-linear relationships between player attributes and ratings. This is appropriate when the relationship isn't linear.

4. Model Evaluation:
   The models will be evaluated using R-squared and MSE to get the accuracy and best fit model .